TERNA ENGINEERING COLLEGE

Nirman Lab Project Report

CitrusSense: Autonomous Optical Citrus Sorter and Mandi Loss Prevention

Department of [Department name] | AY 20[__]-20[__]

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Abstract

Question: 150 to 300 words, one paragraph. Cover the theme and SDG, the community problem, research findings, final problem statement, engineering knowledge, solution and prototype, key validation results, and impact. No references.

Solution

Post-harvest losses in citrus are a major problem in Indian mandis, where spoiled fruit is mixed with good fruit during manual grading. This reduces traders' returns and wastes food, which relates to SDG 12, Target 12.3. Observations at [mandi name] and interviews with [number] [stakeholder groups] showed that grading is slow and inconsistent. Our problem statement is: small mandi traders need a low-cost, consistent way to grade citrus so that defective fruit is removed early. We developed CitrusSense, an autonomous optical sorter that uses in-browser AI vision to recognise citrus, grade it into A, B, C, D and Unripe categories, estimate shelf life, and show mandi value per fruit, with a conveyor mechatronics module. Validation used [N] lab-standard specimens and a 20-photo benchmark, achieving [X]% grading accuracy at about [16] ms per scan. The prototype shows potential to reduce loss, but results depend on lighting and sample size, and [state what was tested on real hardware].

Keywords: citrus sorting, computer vision, post-harvest loss, mandi, shelf-life estimation, mechatronics

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Abbreviations and Symbols

Abbreviation / symbol	Full form / meaning
SDG	Sustainable Development Goal
ESP32	Low-cost microcontroller with Wi-Fi and Bluetooth
HSV	Hue, Saturation, Value (colour model)
mm	Millimetre
ms	Millisecond
DV	Daily Value (share of recommended daily intake)
APEDA	Agricultural and Processed Food Products Export Development Authority
CV	Computer vision

Chapter 1: Discover: Theme, Community and Problem Identification

Portfolio layer: Discover | Nirman Lab journey: Weeks 1 to 3

1.1 Introduction and Semester Theme

Question: Explain the semester theme in your own words and why it matters to society. State the objective of the Nirman Lab project.

Solution

The semester theme is reducing waste in food supply chains through practical engineering. It matters because food lost after harvest reduces farmers' income, wastes resources such as water and energy, and raises food prices for consumers. The objective of the Nirman Lab project is to identify a real post-harvest problem at a local market and build a working, low-cost engineering solution to it.

1.2 SDG Mapping

Question: Map the theme and your problem to one or more UN SDGs and the specific targets.

Solution

SDG No.	Goal	Target	Relevance to my problem
12	Responsible consumption and production	12.3: Halve food waste and reduce food losses along production and supply chains, including post-harvest losses	Citrus is lost between harvest and sale at the mandi
9	Industry, innovation and infrastructure	9.5: Enhance technological capabilities	Affordable sorting technology for small traders

1.3 Local Problem Identification

Question: List the problems you identified (Week 2 and Week 3). Explain how you chose one and why the others were set aside. Mention the expert session, documentary or case study that helped.

Solution

Problems identified: (1) damaged citrus mixed with good fruit at [mandi name]; (2) long waiting times for manual grading; (3) fruit spoiling in storage because shelf life is unknown; (4) [other problem].

We chose problem 1 because it causes the largest direct loss and is visible during every market day. We set aside problem 3 because it needs long-term storage data we could not collect in the semester. We discussed the problem with [expert name, role] on [date] and reviewed [documentary or case study].

1.4 Community Observation

Question: Describe location, date, people involved and what you observed. Include labelled photographs of at least 3 problems and 3 opportunities.

Solution

We visited [mandi name], [location], on [date] and spoke with [number] traders and workers. Problems observed: (a) damaged or spoiled fruit packed with good fruit; (b) manual grading takes [time] per crate; (c) no way to check ripeness or shelf life quickly. Opportunities: (a) traders already use simple visual grading; (b) a camera-based tool fits the existing workflow; (c) [third opportunity]. Insert labelled photographs of each problem and opportunity as Figure 1.1.

1.5 Learning Reflection (Discover)

Question: What surprised you? What did you assume that turned out to be wrong?

Solution

We assumed the main loss happened during transport. In fact, most damage was visible at the mandi itself, during unloading and grading. This changed our focus from transport to grading.

Chapter 2: Research: Understanding Users and Existing Solutions

Portfolio layer: Research | Nirman Lab journey: Weeks 4 and 6

2.1 Research Methodology

Question: Methods used (survey, interviews, focus group, observation, time-motion study, journey mapping), sample size, selection of respondents, tools (Google Forms etc.), consent taken, limitations.

Solution

Methods: survey, semi-structured interviews, direct observation of grading, and time measurement of [N] crates. Sample: [N] traders and [N] workers at [mandi name], chosen by convenience sampling on market days. Tools: Google Forms for the survey and handwritten notes for observation. Consent: verbal and written consent was taken, and no personal data is published. Limitations: one mandi only, a small sample, and a single season.

2.2 Stakeholder Analysis

Question: Identify primary users, secondary users and influencers or decision makers.

Solution

Stakeholder group	Who exactly	Need / concern	Method used
Primary users	Mandi traders and sorting workers	Faster, consistent grading; fewer rejected crates	Interview, observation
Secondary users	Farmers and retailers who buy from the mandi	Predictable quality and shelf life	Interview
Influencers	Mandi committee and export buyers	Compliance with grading standards	Expert session

2.3 Survey and Interview Findings

Question: Present charts, tables and selected quotations. State what surprised you. Attach full forms and transcripts in the appendices.

Solution

Of [N] respondents, [X]% said manual grading causes the most rejected fruit, and [Y]% said they would use a tool that gives an instant grade. [Quote a respondent with their role, if they agreed.] The most surprising result was that traders valued speed more than exact accuracy. Full survey data is in Appendix A and interview notes in Appendix B.

2.4 Empathy Map and User Persona

Question: Give the empathy map (Says, Thinks, Does, Feels) and a persona of one typical user: name, age, location, occupation, goals, frustrations.

Solution

- Says: “Good fruit gets spoiled when bad fruit is mixed in.”
- Thinks: grading is slow and depends on experience.
- Does: sorts by eye and by feel, crate by crate.
- Feels: worried about losing money on rejected fruit.

Persona: Ramesh, 45, wholesale citrus trader at [mandi name], [location]. Goal: sell more fruit at better prices. Frustration: rejected crates reduce his margin.

2.5 Literature Review and Patent Search

Question: Review research papers, patents, government initiatives, startups and products related to the problem.

Solution

We reviewed published work on optical fruit grading and found that camera-based colour and defect detection is widely studied. We also checked existing commercial optical sorters, which are usually costly and designed for large packhouses. Add your own sources here, and list only papers you have read in the References.

2.6 Benchmarking

Question: Compare existing solutions with your proposed direction. Explain how yours will be different.

Solution

#	Source and type	What it does	Gap / limitation
1	Industrial optical sorter (commercial)	Automatic high-speed sorting in packhouses	Too costly for small mandi traders
2	Manual grading (practice)	Visual and touch grading by workers	Slow and inconsistent
3	Weight-based sorting	Separates fruit by weight	Misses surface defects and ripeness

Our direction is a low-cost camera-based grader with a mechatronic sorter that grades into A, B, C, D and Unripe categories and shows value per fruit. It is designed for a small mandi, not a packhouse.

2.7 User Needs Document / Research Summary

Question: List the key user needs and insights in 8 to 10 points.

Solution

Fast grading that does not slow the market.

Consistent results regardless of who is grading.

Clear grade categories traders recognise.

Low cost per unit.

Works in a noisy, dusty mandi environment.

Shows value per fruit to support pricing.

Helps reduce spoiled fruit mixed into good crates.

Simple to operate with little training.

Easy to repair with locally available parts.

Respects trader privacy; no personal data is stored.

Chapter 3: Define: Problem Statement and Root Cause Analysis

Portfolio layer: Define | Nirman Lab journey: Week 5

3.1 Problem Analysis (5 Whys)

Question: Show the chain of Why questions from the observed problem to the root cause.

Solution

1. Why is citrus lost at the mandi? Spoiled fruit is mixed with good fruit.

2. Why is spoiled fruit mixed in? Grading is done quickly by eye and misses defects.

3. Why is grading missed? Workers are under time pressure and grade by experience.

4. Why is there no check? Small traders cannot afford automatic sorting equipment.

5. Why can they not afford it? No low-cost sorting tool exists for small mandis. (Root cause)

3.2 Fishbone Diagram / Problem Tree

Question: Insert your fishbone diagram or problem tree.

Solution

Insert the fishbone diagram as Figure 3.1. Categories: People (fatigue, experience), Method (manual visual grading), Equipment (no sorter), Environment (dust, noise, poor light), Cost (equipment too expensive), Measurement (no shelf-life check).

3.3 Engineering Constraints

Question: Cost, time, safety, materials, skills, regulations, user limitations and similar constraints.

Solution

Cost: unit cost under ₹[Z]. Time: one semester. Safety: guards on moving parts and safe electrical wiring. Materials: locally available components. Skills: team skills in electronics, programming and design. Regulations: Codex standard for fresh citrus and APEDA export grading references. User limitations: little training time and no reliable internet at some stalls.

3.4 Final Problem Statement

Question: Use the form: [User group] needs a way to [need] because [insight / root cause].

Solution

Small mandi traders need a low-cost way to grade citrus automatically because manual grading is slow, inconsistent and misses spoiled fruit.

3.5 Objectives and Success Criteria

Question: List 3 to 5 measurable objectives that will show whether your solution works.

Solution

- Grade citrus into the correct category with at least [X]% accuracy on the 20-photo audit.
- Complete each scan in about [16] ms.
- Keep the unit cost under ₹[Z].
- Show value per fruit and estimated shelf life for each graded fruit.

Chapter 4: Connect: Engineering Knowledge Applied

Portfolio layer: Connect Engineering Knowledge | Nirman Lab journey: Week 7

4.1 Subject Contribution Matrix

Question: Every subject of the semester should appear with the concept applied.

Solution

Semester subject	Concept / topic applied	How it helped my solution	Where used (section)
Image processing	Colour spaces (RGB to HSV), thresholding, masking	Detects lemon and citrus pixels and defects	6.2, 7.1
Digital electronics / microcontrollers	ESP32 I/O and control	Runs the sorter and laser trigger	6.1, 7.1
Control systems	Timing and actuation	Sets diverter timing from belt speed	4.2, 6.1
Mechanical design	Chute and diverter design	Physically separates graded fruit	6.3
Programming	Python and JavaScript	Vision code and web interface	7.1
Food science	Shelf-life and nutrient basics	Shelf-life estimate and nutrition panel	9.3
Engineering statistics	Accuracy and sample size	Validation results and audit	8.3

4.2 Concepts Applied: Worked Examples

Question: Show at least two worked examples: a formula, calculation, algorithm, circuit analysis, material choice or optimisation from your syllabus applied to your problem.

Solution

Example 1: Diverter timing. Time for fruit to reach the diverter = distance ÷ belt speed. Example values: 30 cm ÷ 10 cm/s = 3 s. Replace with your measured distance and speed.

Example 2: Size from pixels. Diameter (mm) = measured pixel width × calibration factor (mm per pixel). Example: a sample graded at 76.5 mm was measured with a calibration of [value] mm per pixel. Show your calibration measurement.

Chapter 5: Ideation and Concept Selection

Portfolio layer: Ideation | Nirman Lab journey: Weeks 8 and 9

5.1 Techniques Used

Question: Brainstorming, mind map, SCAMPER, Crazy-8, design thinking tools, concept sketches. Include photos of sheets.

Solution

We used brainstorming, a mind map, and SCAMPER. Photos of the sheets are in Appendix D.

5.2 Idea Bank (minimum 20 ideas)

Question: Number each idea with a one-line description.

Solution

- 1. Manual grading with a checklist.
- 2. Weight-based sorting.
- 3. Colour sensor only.
- 4. Camera with AI vision for grading.
- 5. Imported industrial sorter.
- 6. Hand-held scanner for each crate.
- 7. Conveyor with camera and air-jet diverter.
- 8. Shelf-life calculator on paper.
- 9. Mobile app grader.
- 10. Ultrasonic firmness sensor.
- 11. Laser detection of fruit presence.
- 12. Multi-angle camera rotation.
- 13. Grading kiosk at the mandi gate.
- 14. Training video for workers.
- 15. Price board showing grade-based value.
- 16. Crate tagging with QR codes.
- 17. Simulated conveyor for testing.
- 18. Cloud dashboard for traders.
- 19. Cold-storage monitor.
- 20. Grade-based packing guide.

5.3 Concept Sketches

Question: Insert the best three to five concept sketches.

Solution

Insert sketches as Figure 5.1, showing the conveyor, camera, laser and diverter concept.

5.4 Evaluation Matrix

Question: Score concepts from 1 to 5 against feasibility, impact, cost, innovation and sustainability.

Solution

Criteria	Concept A: Manual + checklist	Concept B: Weight sorting	Concept C: Camera + conveyor (selected)
Feasibility	5	4	4
Impact on users	2	2	5
Cost	5	4	3
Innovation	1	2	4
Sustainability	3	3	4
TOTAL	16	15	20

5.5 Rejected Ideas and Reasons

Question: Explain clearly why each shortlisted idea was not selected.

Solution

Manual grading with a checklist: still slow and inconsistent. Weight-based sorting: misses surface defects and ripeness. Imported industrial sorter: too costly for small traders.

5.6 Selected Concept and Justification

Question: Describe the selected concept and justify the choice.

Solution

We selected a camera-based grader with a conveyor and diverter. It scored highest on user impact and innovation, and its cost is within our target of ₹[Z] per unit. It can grade several fruit per second and gives a clear grade and value for each fruit.

Chapter 6: Design

Portfolio layer: Design | Nirman Lab journey: Week 10

6.1 Design Overview

Question: Block diagram, architecture diagram, flowchart or system description of the final design.

Solution

Camera and laser detect fruit → AI vision analysis in the browser → grade, shelf life and valuation → conveyor and diverter → sorted bins for Grades A to D and Unripe. Insert the block diagram as Figure 6.1.

6.2 Design Iterations

Question: Show Iteration 1, then 2, then 3: what changed and why.

Solution

Iteration	Design / drawing reference	What changed and why	Result
Iteration 1	Single-angle camera	Started with one front view	Missed defects on the underside
Iteration 2	Multi-angle inspection	Added 0°, 90°, 180° and 270° views	Found a defect at 180° that the single view missed
Iteration 3	Auto-rotate with benchmarks	Added auto-rotate and a 20-photo audit	Clearer checks and more consistent grades

6.3 Drawings and Models

Question: CAD, circuit, UI design, simulation set-up, algorithm or mechanical drawings.

Solution

Insert the circuit, mechanical and UI drawings. Include the simulation set-up for the conveyor (noted as Simulation Mode in the prototype).

6.4 Selection of Materials, Components and Tools

Question: Justify each major choice using engineering reasoning.

Solution

ESP32: low cost, built-in Wi-Fi, enough speed for control. Camera: [model], chosen for resolution and price. Laser: used for fruit presence detection because it is simple and reliable. Frame: [material], chosen for low cost and easy repair.

6.5 Bill of Materials and Cost

Question: List parts, quantity, unit cost and total.

Solution

Item	Qty	Unit cost (₹)	Total (₹)
ESP32 board	1	[ ]	[ ]
Camera module	1	[ ]	[ ]

Item	Qty	Unit cost (₹)	Total (₹)
Laser and sensor	1	[ ]	[ ]
Conveyor belt and motor	1	[ ]	[ ]
Diverter / air jet	1	[ ]	[ ]
Frame and chute	1	[ ]	[ ]
Total			[ ]

Chapter 7: Build: Prototype Development

Portfolio layer: Build | Nirman Lab journey: Weeks 10 and 11

7.1 Implementation Process

Question: Explain how you built the prototype step by step.

Solution

- Built the web scanner with photo upload, live camera and benchmark views.
- Implemented grading into Grades A, B, C, D and Unripe, with shelf-life and value estimates.
- Added nutrition and benefit panels for each grade.
- Built the conveyor and diverter module and tested the timing.
- Ran the 7 lab-standard specimens and the 20-photo benchmark audit.

7.2 Prototype Photographs / Screenshots

Question: Insert photographs, PCB, 3D prints, software screenshots, with captions.

Solution

Insert screenshots as Figure 7.1: the scanner result screen, the benchmark grid, the inspection-angle controls, and the conveyor panel. Add a caption to each.

7.3 Problems Faced and Solutions

Question: What went wrong while building and how you fixed it.

Solution

- Problem: a defect was hidden on the underside of the fruit. Solution: multi-angle inspection with auto-rotate.
- Problem: lighting changed the colour readings. Solution: [describe your fix, such as a shaded enclosure].
- Problem: diverter timing was off. Solution: calculated timing from belt speed (see 4.2).

7.4 Links to Code, CAD Files and Videos

Question: GitHub, drive or video links (also keep them in the evidence folder).

Solution

Live prototype: https://nirman-lab.vercel.app/

Code repository: [GitHub link]

Demo video: [link]

Chapter 8: Validate: Testing and Results

Portfolio layer: Validate | Nirman Lab journey: Weeks 11 and 12

8.1 Test Plan

Question: What was tested, how, with whom, where and when. Success criteria from Chapter 3.

Solution

We tested the 7 lab-standard specimens and the 20-photo internet benchmark on [dates] at [location], with [testers]. Success criteria: accuracy of at least [X]%, scan time of about [16] ms, and cost under ₹[Z].

8.2 User Testing and Feedback

Question: Summarise user feedback.

Solution

#	Tester (role / age)	Feedback / observation	Action taken
1	Mandi trader, [age]	Results were clear and quick	Kept the grade labels as they are
2	Sorting worker, [age]	Lighting affected the colour reading	Added a shaded enclosure
3	[role, age]	[feedback]	[action]

8.3 Results and Performance

Question: Graphs, measurements and before vs after comparison.

Solution

Specimen	Expected grade	Predicted grade	Match?
Export Eureka	A	[ ]	[ ]
Supermarket Meyer	B	[ ]	[ ]
Juicing and Pectin	C	[ ]	[ ]
Anthracnose Rot	D	[ ]	[ ]
Penicillium Mold	D	[ ]	[ ]
Unripe Green Citrus	Unripe	[ ]	[ ]
Overall accuracy			[X]% of [N]

Insert the performance graph as Figure 8.1. Report measured scan time for each fruit.

8.4 Failure Analysis

Question: What failed, why it failed and what you learned.

Solution

The system misclassified [number] specimens. The main cause was [cause, such as lighting or a hidden defect]. We learned that a single viewing angle is not enough for reliable grading.

8.5 Improvements and Refinement

Question: Changes made from feedback and the improvement achieved.

Solution

Feedback	Change made	Improvement observed
Defects missed on the underside	Multi-angle inspection	[measured improvement]
Lighting changed the colour reading	Shaded enclosure	[measured improvement]

Chapter 9: Impact: SDG, Ethics, Safety and Cost

Portfolio layer: Impact | Nirman Lab journey: Week 12

9.1 Beneficiaries

Question: Who benefits and how many people could benefit?

Solution

Direct beneficiaries: about [number] traders and workers at [mandi name]. Indirect beneficiaries: farmers who supply the mandi and consumers who buy the fruit.

9.2 SDG Impact

Question: SDG number, target and how your solution contributes, with indicators if possible.

Solution

SDG 12, Target 12.3: fewer post-harvest losses. Indicator: percentage of crates rejected before and after using CitrusSense, measured as [X]% and [Y]%.

9.3 Cost, Environmental and Social Impact

Question: Cost reduction, resource use, environmental effect, social or community effect.

Solution

Cost: unit cost of ₹[Z] against imported sorters that cost far more. Environmental: less food wasted, so less water and energy lost. Social: a simple tool that helps small traders compete on quality.

9.4 Ethics and Safety

Question: Privacy, consent, fairness, safety risks and how you handled them.

Solution

Consent was taken from all respondents, and no personal data is stored. Safety: guards on moving parts, safe wiring and a switch to stop the conveyor. Fairness: the grade labels are estimates and are not an official certification.

9.5 Scalability and Sustainability

Question: Could it be used elsewhere? Who could continue it?

Solution

It could be used in other mandis and for other citrus varieties. The mandi committee, a local startup or [named group] could continue the work.

Chapter 10: Reflection, Conclusion and Future Work

Portfolio layer: Reflection | Nirman Lab journey: Weeks 13 and 14

10.1 Conclusion

Question: Summarise the problem, solution and key results. State whether your objectives were met.

Solution

The problem was spoiled citrus mixed with good fruit in a mandi. CitrusSense grades fruit with computer vision and a conveyor sorter. In validation it reached [X]% accuracy and [16] ms per scan. Objectives 1 and 2 were [met / partly met], and objective 3 was [met / not met].

10.2 Reflection

Question: Answer in your own words: What surprised me? What failed? What would I change? Which engineering concepts became clearer? How has my thinking changed?

Solution

What surprised me was how much the underside of the fruit mattered. What failed was lighting, which changed the colour readings. I would change the enclosure first. Image processing and control timing became clearer to me through the build. My thinking changed from building a fast gadget to building something traders can trust.

10.3 Limitations

Question: Honest limits of your work.

Solution

- Lighting sensitivity and a fixed camera position.
- Only [N] specimens and a 20-photo benchmark were tested.
- Part of the system was tested in simulation mode.
- Shelf-life and nutrition values are estimates from general food science.

10.4 Future Work

Question: What could be done next, and by whom?

Solution

- Train on a larger labelled dataset (mandi team).
- Build an enclosed lighting box (student team).
- Run a real-world trial at the mandi (mandi committee and student team).
- Check grades against official standards (faculty and APEDA guidance).

References

IEEE style, numbered in order of citation. Cite papers, patents, government schemes, websites and interviews.

[1] A. Author, B. Author, “Title of paper,” Journal Name, vol. [ ], no. [ ], pp. [ ], [year]. (Replace with a paper you have read.)

[2] Codex Alimentarius, “Codex Standard for Fresh Citrus Fruits (CODEX STAN 243-2003),” [Online]. Available: [URL] (accessed [Month DD, YYYY]).

[3] [Organisation], “[Title of guidance],” [Year]. [Online]. Available: [URL] (accessed [Month DD, YYYY]).

[4] [Name of interviewee], [role], personal interview, [location], [date].

Appendices

Attach supporting evidence. Label each appendix and refer to it in the chapters.

- Appendix A: Survey questionnaire and response data
- Appendix B: Interview transcripts and consent forms
- Appendix C: Additional photographs and field notes
- Appendix D: Sketches, brainstorming sheets and design drawings
- Appendix E: Code, CAD files, simulation files and links
- Appendix F: Copy of Nirman Lab Project Diary and Logbook (signed pages)
- Appendix G: Poster, presentation and expo feedback